JEE Mathematics: Quick Review with Grading and Practice Map - Binomial Theorem (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (105 questions; a question that appears in two of these is counted once): High = 10 or more questions, Medium = 1 to 9 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction

Question 01

What do monomial, binomial, trinomial and multinomial expressions mean?

Model Answer

Monomial: 1 term; binomial: 2 terms; trinomial: 3 terms; multinomial: more than 2 terms.

Ex: $a + \dfrac{b}{x}$ is binomial; $x + y + z$ is trinomial.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

Are you able to draw Pascal's triangle and what is its use?

Model Answer

Row n gives the coefficients of $(x + y)^n$; each entry is the sum of the two entries above it, and each row starts and ends with 1.

Ex: row 5 is $1,\ 5,\ 10,\ 10,\ 5,\ 1$ (as $4 + 6 = 10$ in the row above).

Grade - Low

Practice map, main concept: none

Also used in: none

Newton's Binomial Theorem

Question 03

Can you write down the binomial theorem?

Model Answer

$$(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + \cdots + {}^nC_r y^r x^{n-r} + \cdots + {}^nC_n y^n, \quad n \in \mathbb{N}$$

Ex: $(x + 2y)^5 = x^5 + 10x^4 y + 40x^3 y^2 + 80x^2 y^3 + 80xy^4 + 32y^5$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 04
What is the number of terms in the expansion of $(x+y)^n$?

Model Answer
$n+1$ terms (one more than the index); in every term the powers of x and y add up to n.
Ex: $(x+y)^9$ has 10 terms.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 05
What is the difference between coefficients and binomial coefficients?

Model Answer
Binomial coefficients are only ${}^nC_0, {}^nC_1, \ldots$; the coefficient of a term also includes the numbers attached to the variables.
Ex: in $(x-2y)^5$ the 2nd term is $-10x^4y$: binomial coefficient ${}^5C_1 = 5$, coefficient -10.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 06
Can you find the coefficient and the binomial coefficient of any term?

Model Answer
Write $T_{r+1} = {}^nC_r\, x^{n-r} y^r$; the binomial coefficient is ${}^nC_r$, the coefficient is ${}^nC_r$ times the numerical factors of x and y.
Ex: $(2x-3)^7$, term in x^4 ($r=3$): binomial coefficient 35, coefficient $35 \cdot 2^4(-3)^3 = -15120$.

Grade - Medium
Practice map, main concept: Sheet: J-I 11
Also used in: Sheet: J-I 5-8, 35; J-II 41; J-M 3, 14, 24; C-I 9 | DPP: 1(11) | Question Bank: 275, 296, 304

Question 07
How to find the sum of coefficients and the sum of binomial coefficients?

Model Answer
Sum of coefficients: put every variable = 1. Sum of binomial coefficients $= {}^nC_0 + \cdots + {}^nC_n = 2^n$.
Ex: $(2x+y)^6$: sum of coefficients $3^6 = 729$, of binomial coefficients $2^6 = 64$.

Grade - Medium
Practice map, main concept: Sheet: J-II 22, 27, 32; J-M 37, 49; C-II 14, 19-20
Also used in: Sheet: J-I 26-27, 29, 33; J-II 26; J-M 46, 68; C-I 2, 6, 16; C-II 7-8, 15-16 | DPP: 1(3-4); 3(10, 12); 4(14, 16) | Question Bank: 305

Question 08
Write the expansions of $(1+x)^n$ and $(1-x)^n$.

Model Answer

$$(1+x)^n = \sum_{r=0}^{n} {}^nC_r x^r, \qquad (1-x)^n = \sum_{r=0}^{n} (-1)^r\, {}^nC_r x^r$$

Ex: $(1-x)^4 = 1 - 4x + 6x^2 - 4x^3 + x^4$.

Grade - Low
Practice map, main concept: none

Also used in: none

Question 09

What is the coefficient of x^r in the expansion of $(1+x)^n$?

Model Answer

It is ${}^nC_r$ (and 0 if $r > n$).

Ex: coefficient of x^3 in $(1+x)^{10}$ is ${}^{10}C_3 = 120$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 11, 15

Question 10

Write $(x+y)^n$ as a sigma of terms.

Model Answer

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^{n-r}y^r = \sum_{r=0}^{n} {}^nC_r\, x^r y^{n-r}$$

Ex: $\displaystyle\sum_{r=0}^{7} {}^7C_r 2^{7-r}3^r = (2+3)^7 = 78125.$

Grade - Low

Practice map, main concept: none

Also used in: none

General Term

Question 11

Do you know the formula of the general term in the expansion of $(x+y)^n$?

Model Answer

$$T_{r+1} = {}^nC_r\, x^{n-r}y^r \qquad (r = 0, 1, \ldots, n)$$

Ex: 10th term of $\left(2x^2 + \dfrac{1}{x}\right)^{12}$: $T_{10} = {}^{12}C_9(2x^2)^3\left(\dfrac{1}{x}\right)^9 = \dfrac{1760}{x^3}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 10; J-M 24; C-I 14; C-II 11 | DPP: 1(7, 10, 15) | Question Bank: 304

Also used in: Sheet: J-I 1, 9; J-II 2; J-M 12, 61, 71 | Question Bank: 295

Question 12

How to find any term from the end in a binomial expansion?

Model Answer

The kth term from the end is the $(n-k+2)$th term from the beginning, T_{n-k+2}.

Ex: 4th from the end in $\left(2x - \dfrac{1}{x^2}\right)^{10}$ is $T_8 = {}^{10}C_7(2x)^3\left(-\dfrac{1}{x^2}\right)^7 = -960x^{-11}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 9; J-M 26, 35 | Question Bank: 281, 285

Also used in: none

Question 13

How to find the coefficient of any power of x (or the term independent of x) in a binomial expansion?

Model Answer

Collect the power of x in T_{r+1}, equate it to the required power (0 for the independent term), solve for r; r must be an integer with $0 \le r \le n$, else the coefficient is 0.

Ex: $\left(x^2 + \frac{2}{x}\right)^6$: $T_{r+1} = {}^6C_r 2^r x^{12-3r}$; x^3: $r = 3$, coefficient 160; independent term: $r = 4$, 240; x^2: r not an integer, coefficient 0.

Grade - High

Practice map, main concept: Sheet: J-I 8, 14; J-II 2-3, 8, 33, 35, 41; J-M 6, 11-12, 14, 17, 21, 25, 27, 30, 32, 40, 42-43, 46, 52, 63, 65, 67, 69; J-A 7; C-I 2, 13 | DPP: 1(1, 3, 8, 11) | Question Bank: 262, 265, 282, 291, 295-296, 301, 305

Also used in: Sheet: J-I 17; J-II 5-6, 36; J-M 15, 59; C-I 1, 12, 15; C-II 1 | DPP: 1(5-6, 10) | Question Bank: 257, 297

Question 14

How to find the number of rational terms in a binomial expansion?

Model Answer

Write T_{r+1} with fractional powers; a term is rational (integral) when every power is an integer. Find the r, $0 \le r \le n$, satisfying all the conditions (usually r a multiple of the lcm of the denominators).

Ex: $\left(5^{1/3} + 7^{1/4}\right)^{60}$: r multiple of 12, $r = 0, 12, \ldots, 60$: 6 terms. $\left(5^{1/2} + 7^{1/4}\right)^{99}$: none.

Grade - Medium

Practice map, main concept: Sheet: J-M 45, 56, 61 | DPP: 1(9, 14, 16)

Also used in: none

Question 15

How to handle the ratio of consecutive terms or coefficients (three consecutive coefficients in a given ratio or in AP)?

Model Answer

$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \cdot \frac{y}{x}$ and $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$. For consecutive coefficients $a : b : c$ form two equations in n, r and solve (for AP use $2b = a + c$).

Ex: coefficients of three consecutive terms of $(1+x)^n$ in ratio $1 : 3 : 5$: $n = 4r - 1$, $r = 2$, $n = 7$ (coefficients $7, 21, 35$).

Grade - High

Practice map, main concept: Sheet: J-I 1, 7; J-II 1; J-M 3, 18, 29, 36, 51, 62, 64, 71; J-A 1; C-I 15 | Question Bank: 256, 260, 275

Also used in: Sheet: J-M 50; C-II 5 | Question Bank: 267

Question 16

How to solve equations involving binomial coefficients (${}^nC_x = {}^nC_y$, Pascal's rule, equal coefficients of two terms)?

Model Answer

${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$; ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$. Reject the root that makes an index negative or exceed n.

Ex: coefficients of $(r+2)$th and $3r$th terms of $(1+x)^{2n}$ equal: ${}^{2n}C_{r+1} = {}^{2n}C_{3r-1} \Rightarrow r = 1$ or $(r+1) + (3r-1) = 2n$, i.e. $r = \frac{n}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 5; J-II 28; J-M 50; C-I 8, 10 | DPP: 3(11) | Question Bank: 267

Also used in: Sheet: J-I 16; J-II 1; J-M 64 | DPP: 3(9)

Question 17

How to find the maximum / least value of a term (or coefficient) that contains a parameter such as $\sin\theta$ or x?

Model Answer

Find the term independent of x (or the required term) as a function of the parameter, then use the bounds ($\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta \le \frac{1}{2}$, AM-GM, or calculus).

Ex: $\left(x\sin\theta + \dfrac{\cos\theta}{x}\right)^{10}$: independent term ${}^{10}C_5 \sin^5\theta\cos^5\theta = \dfrac{252}{32}\sin^5 2\theta$, greatest value $\dfrac{63}{8}$.

Grade - Medium

Practice map, main concept: Question Bank: 269, 272

Also used in: none

Middle Term

Question 18

How to find the position of the middle term in a binomial expansion?

Model Answer

n even: one middle term $T_{\frac{n}{2}+1}$. n odd: two middle terms $T_{\frac{n+1}{2}}$ and $T_{\frac{n+3}{2}}$.

Ex: $n = 30$: T_{16}; $n = 49$: T_{25} and T_{26}.

Grade - Medium

Practice map, main concept: Sheet: J-I 6; J-II 36

Also used in: Sheet: J-M 5 | Question Bank: 276

Question 19

Do you know that the binomial coefficient of the middle term is the greatest binomial coefficient?

Model Answer

Greatest of ${}^nC_r$ is ${}^nC_{n/2}$ (n even) or ${}^nC_{\frac{n-1}{2}} = {}^nC_{\frac{n+1}{2}}$ (n odd); ${}^nC_r = {}^nC_s \Rightarrow r = s$ or $r + s = n$.

Ex: greatest binomial coefficient of $(1+x)^{10}$ is ${}^{10}C_5 = 252$; of $(x+y)^9$ it is ${}^9C_4 = {}^9C_5 = 126$.

Grade - Low

Practice map, main concept: Sheet: J-I 3; C-II 16 | DPP: 1(4); 4(13)

Also used in: none

Question 20

How to find the coefficient of any power of x in a product of two or more binomial expansions?

Model Answer

Take the general term of each factor; the coefficient is the sum of products of coefficients whose powers add up to the required power.

Ex: coefficient of x^3 in $(2+3x-5x^2)(1+x)^{10}$: $2\cdot{}^{10}C_3 + 3\cdot{}^{10}C_2 - 5\cdot{}^{10}C_1 = 325$.

Grade - Medium

Practice map, main concept: Sheet: J-I 15-16, 18; J-II 6-7, 39; J-M 15, 34, 58, 60; J-A 2; C-I 5, 9, 16; C-II 9 | DPP: 2(1-2, 10-11); 4(12) | Question Bank: 261, 297

Also used in: Sheet: J-I 12; J-II 22; J-M 53 | DPP: 3(2) | Question Bank: 263

Numerically Greatest Term (NGT)

Question 21

How to find the NGT (numerically greatest term) in a binomial expansion?

Model Answer

For $(a+b)^n$ find $p = \dfrac{n+1}{1+\left|\frac{a}{b}\right|}$. If p is not an integer, NGT is $T_{[p]+1}$; if p is an integer, T_p and T_{p+1} are equal and both greatest.

Ex: $(3-2x)^9$: $x=\frac{1}{2}$: $p=2.5 \Rightarrow T_3$; $x=1$: $p=4 \Rightarrow T_4 = T_5$ (in magnitude).

Grade - Low

Practice map, main concept: Sheet: J-II 9; C-II 15 | DPP: 1(12); 2(12)

Also used in: Sheet: J-I 23; J-II 10

Question 22

If the position of the NGT is given, can you write the condition on p?

Model Answer

T_k is the only NGT $\Leftrightarrow k-1<p<k$; T_k and T_{k+1} both NGT $\Leftrightarrow p=k$. Then solve for x or n.

Ex: T_4 is the NGT of $(1+x)^{10}$ $(x>0)$: $3 < \dfrac{11x}{1+x} < 4 \Rightarrow \dfrac{3}{8} < x < \dfrac{4}{7}$.

Grade - Low

Practice map, main concept: Sheet: J-I 23; J-II 10-11

Also used in: none

Remainder Based Problems

Question 23

How to convert the base to the nearest multiple of the divisor to find a remainder?

Model Answer

Write the base as (multiple of divisor) $\pm$ small number, expand by the binomial theorem; every term except the last is divisible by the divisor. If the remainder is negative add the divisor.

Ex: $3^{2022} = 9^{1011} = (10-1)^{1011} \equiv (-1)^{1011} = -1 \equiv 4 \pmod 5$.

Grade - High

Practice map, main concept: Sheet: J-I 21-22; J-M 16, 20, 22-23, 33, 39, 47-48, 55, 73; C-I 7 | DPP: 2(3, 7, 9) | Question Bank: 284, 292-293, 298, 300, 302-303, 306-307

Also used in: Sheet: J-I 19; J-M 28 | DPP: 2(4)

Question 24

What to do when the base is very small compared to the divisor?

Model Answer

Raise the base to a suitable power first (e.g. a^k) so that it becomes close to a multiple of the divisor, then expand.

Ex: $2^{100} = 2 \cdot (2^3)^{33} = 2(7+1)^{33} \equiv 2 \pmod 7$.

Grade - Medium

Practice map, main concept: none

Also used in: Question Bank: 292

Question 25

How many terms do we have to consider when finding a remainder?

Model Answer

If the divisor is d and base $= kd \pm 1$, one term suffices; if the divisor is d^2 keep the first two terms: $(1+kd)^n \equiv 1 + nkd \pmod{d^2}$; in general keep terms until the power of d is as large as the divisor needs.

Ex: $9^n - 8n - 1 = (1+8)^n - 8n - 1 = 64(\dots)$, divisible by 64.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 26

What to keep in mind while finding the unit digit / last two digits / last three digits?

Model Answer

Last 1, 2, 3 digits = remainder on division by $10, 100, 1000$. Write the base as $10k \pm 1$ or (multiple of 10) ± 1 and keep the first 2 or 3 terms. Write the answer with leading zeros (e.g. 01).

Ex: $21^{100} = (1+20)^{100} \equiv 1 + 100 \cdot 20 \equiv 1 \pmod{100}$: last two digits 01.

Grade - Low

Practice map, main concept: Sheet: J-I 19-20 | DPP: 2(5, 8); 3(10)

Also used in: none

Question 27

What is the condition for $a^n + b^n$ to be divisible by $a + b$?

Model Answer

$a^n + b^n$ is divisible by $a + b$ when n is an odd natural number.

Ex: $11^{25} + 12^{25}$ is divisible by $11 + 12 = 23$.

Grade - Low

Practice map, main concept: DPP: 2(4, 6)

Also used in: none

Question 28

What is the condition for $a^n - b^n$ to be divisible by $a + b$?

Model Answer

$a^n - b^n$ is divisible by $a + b$ when n is an even natural number.

Ex: $8^6 - 3^6 = 261415$ is divisible by $8 + 3 = 11$.

Grade - Medium

Practice map, main concept: Question Bank: 266

Also used in: none

Question 29

What is the condition for $a^n - b^n$ to be divisible by $a - b$?

Model Answer

$a^n - b^n$ is divisible by $a - b$ for every natural number n.

Ex: $13^9 - 4^9$ is divisible by $13 - 4 = 9$.

Grade - Medium

Practice map, main concept: Sheet: J-M 28, 31 | Question Bank: 283

Also used in: none

Integral Part and Fractional Part Based Concept

Question 30

How to prove that the integral part of a binomial expansion is odd / even?

Model Answer

Let $R = (a+\sqrt{b})^n = I + f$ and $G = (a-\sqrt{b})^n$ with $0 < G < 1$. Then $R+G$ is an integer (even or odd as the terms show) and $f + G = 1$, so $I = (R+G) - 1$.

Ex: $(9+4\sqrt{5})^n$, $G = (9-4\sqrt{5})^n \in (0,1)$, $R+G = 2(\dots)$ even, so I is odd.

Grade - Low

Practice map, main concept: Sheet: J-II 14

Also used in: Sheet: J-II 38

Question 31

How to find the integral part and fractional part of a binomial expansion?

Model Answer

With $R = I + f$ and the conjugate G: $f = 1 - G$ (if $0 < G < 1$, $I = R + G - 1$) or $f = -G$ (if $-1 < G < 0$, $I = R + G$), so $f(1-f)$-type products reduce to $R \cdot G = (a^2 - b)^n$.

Ex: $(5+2\sqrt{6})^n = I + f$, $G = (5-2\sqrt{6})^n = 1 - f$ and $RG = 1$, so $(I+f)(1-f) = 1$; for $n = 1$: $I = 9$.

Grade - Medium

Practice map, main concept: Sheet: J-I 24; J-II 15-17, 38; J-M 8; C-II 4, 10 | DPP: 3(4-6) | Question Bank: 278

Also used in: Sheet: J-II 14, 37; J-M 39; C-II 14

Question 32

How to handle $(x+a)^n \pm (x-a)^n$ (even / odd terms only, degree of the polynomial)?

Model Answer

$(x+a)^n + (x-a)^n = 2\left[{}^nC_0 x^n + {}^nC_2 x^{n-2}a^2 + \dots\right]$ (odd r cancel) and the difference keeps the odd r. Take the highest power from the last surviving term.

Ex: $(1+\sqrt{2})^4 + (1-\sqrt{2})^4 = 2(1 + 6\cdot 2 + 4) = 34$.

Grade - Medium

Practice map, main concept: Sheet: J-I 4, 34 | DPP: 1(13) | Question Bank: 257, 268

Also used in: none

Multinomial Theorem

Question 33

Write the general term of a multinomial expansion.

Model Answer

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{r_1!\, r_2! \cdots r_k!}\, x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k}, \qquad r_1 + \dots + r_k = n$$

Ex: coefficient of $x^2y^3z^4$ in $(x+y+z)^9$ is $\dfrac{9!}{2!\,3!\,4!} = 1260$.

Grade - Low

Practice map, main concept: Sheet: J-II 12 | DPP: 3(1, 16)

Also used in: Sheet: J-II 34

Question 34
How to find the number of dissimilar terms in a multinomial expansion?

Model Answer
Number of non-negative solutions of $r_1 + \cdots + r_k = n$: ${}^{n+k-1}C_{k-1}$.
Ex: $(a+b+c)^{10}$ has ${}^{12}C_2 = 66$ terms.

Grade - Low
Practice map, main concept: Sheet: J-I 2; J-II 34
Also used in: none

Question 35
How to use the general term to find the coefficient of a power of x in a multinomial expansion?

Model Answer
Write $\dfrac{n!}{r_1!r_2!r_3!}(\cdot)^{r_1}(\cdot)^{r_2}(\cdot)^{r_3}$, impose $r_1 + r_2 + r_3 = n$ and (power of x) = required power, list all non-negative integer solutions and add.
Ex: x^2 in $(1+x+x^2)^4$: $(r_1, r_2, r_3) = (2,2,0), (3,0,1)$ give $6 + 4 = 10$.

Grade - Medium
Practice map, main concept: Sheet: J-II 4, 13; J-M 41, 44 | Question Bank: 287, 290
Also used in: Sheet: J-II 12 | DPP: 3(1, 16)

Question 36
Try to factorise the base before applying the multinomial theorem.

Model Answer
Factorise (or sum a GP) so that each factor is a binomial, then use the product rule of item 20.
Ex: $x^2 - x - 2 = (x-2)(x+1)$, so $(x^2 - x - 2)^5 = (x-2)^5(x+1)^5$ and the coefficient of x^5 is -81.

Grade - Medium
Practice map, main concept: Sheet: J-I 17; J-M 53 | DPP: 1(5); 3(2-3) | Question Bank: 263
Also used in: Sheet: J-II 13

Question 37
How to count the distinct terms in a sum of several expansions?

Model Answer
List the set of powers of x present in each expansion, take their union and count it (a power common to two expansions is one term).
Ex: $(1+x)^4 + (1+x^2)^3$: powers $\{0,1,2,3,4\} \cup \{0,2,4,6\} = \{0,1,2,3,4,6\}$, so 6 terms.

Grade - Low
Practice map, main concept: DPP: 1(17)
Also used in: none

Summation of Series Type 1

Question 38
Write $(x+y)^n$ as a sigma of terms (type 1 form).

Model Answer

$$\sum_{r=0}^{n} {}^{n}C_r \, x^{n-r} y^r = (x+y)^n$$

Ex: $\sum_{r=0}^{7} {}^{7}C_r 2^r = 3^7$.

Grade - Low

Practice map, main concept: Sheet: C-I 1, 3; C-II 2-3 | DPP: 3(12)
Also used in: Sheet: C-II 12 | DPP: 4(6)

Question 39
How to identify the values of x, y and n if a series is a type 1 question?

Model Answer
Write each term as ${}^{n}C_r \, x^{n-r} y^r$: the powers attached to r give y, the remaining power $(n-r)$ gives x ($x = 1$ if absent), n is the upper index.
Ex: $\sum_{r=0}^{6} {}^{6}C_r \frac{2^r}{3^r} = \left(1+\frac{2}{3}\right)^6 = \left(\frac{5}{3}\right)^6$.

Grade - Medium

Practice map, main concept: Sheet: J-II 19 | Question Bank: 294
Also used in: none

Question 40
What to keep in mind while solving type 1 questions?

Model Answer
Check the limits of r (if the sum starts at $r = 1$ subtract the $r = 0$ term), the sign $(-1)^r$ (makes y negative), and powers like $4^r = (2^2)^r$.
Ex: $\sum_{r=1}^{n} {}^{n}C_r 2^r = 3^n - 1$.

Grade - Low

Practice map, main concept: none
Also used in: none

Summation of Series Type 2

Question 41
When to apply type 2 and which formula to use?

Model Answer
When the upper suffix varies and the lower one is fixed: $\sum_{k=r}^{m} {}^{k}C_r = {}^{m+1}C_{r+1}$ (hockey stick). Also coefficient of x^r in $(1+x)^r + \cdots + (1+x)^m$.
Ex: ${}^{10}C_{10} + {}^{11}C_{10} + \cdots + {}^{25}C_{10} = {}^{26}C_{11}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 29; J-M 4; J-A 4 | DPP: 4(1, 5, 7) | Question Bank: 271
Also used in: Sheet: J-M 66

Question 42
How to start combining two terms in type 2 questions?

Model Answer
Use Pascal's rule ${}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$ on the first two terms, then on the result and the next term, and so on.
Ex: ${}^{11}C_{11} + {}^{11}C_{10} = {}^{12}C_{11}$, then ${}^{12}C_{11} + {}^{12}C_{10} = {}^{13}C_{11}$.

Grade - Low
Practice map, main concept: Sheet: J-II 31 | DPP: 3(9, 15); 4(8)
Also used in: none

Question 43
How to rewrite 1 (one) as another binomial coefficient in type 2 questions?

Model Answer
Write $1 = {}^{k}C_k = {}^{k+1}C_{k+1}$ with the suffix that matches the first term, so that Pascal's rule can start.
Ex: $1 + {}^{11}C_{10} = {}^{11}C_{11} + {}^{11}C_{10} = {}^{12}C_{11} = 12$.

Grade - Low
Practice map, main concept: none
Also used in: none

Summation of Series Type 3

Question 44
What to keep in mind if the sign changes alternately in a series involving binomial coefficients?

Model Answer
Put $x = -1$: $\sum(-1)^r {}^{n}C_r = 0$. For a partial sum $\displaystyle\sum_{r=0}^{k}(-1)^r {}^{n}C_r = (-1)^k {}^{n-1}C_k$.
Ex: ${}^{20}C_0 - {}^{20}C_1 + \cdots + {}^{20}C_{10} = {}^{19}C_{10} = \frac{1}{2}\,{}^{20}C_{10} = 92378$.

Grade - Medium
Practice map, main concept: Sheet: J-I 31; J-M 38 | DPP: 4(2) | Question Bank: 286
Also used in: Sheet: C-II 3, 18, 20 | DPP: 4(4)

Question 45
How to find the sum of coefficients of even powers of x in an expansion?

Model Answer
For $f(x) = \sum a_r x^r$: $a_0 + a_2 + a_4 + \cdots = \dfrac{f(1) + f(-1)}{2}$.
Ex: $(1 - x + x^2)^n$: $a_0 + a_2 + \cdots = \dfrac{1 + 3^n}{2}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 28, 32; C-I 4; C-II 13, 18 | DPP: 3(8); 4(10) | Question Bank: 273, 289
Also used in: Sheet: C-II 19

Question 46
How to find the sum of coefficients of odd powers of x in an expansion?

Model Answer
$a_1 + a_3 + a_5 + \cdots = \dfrac{f(1) - f(-1)}{2}$.
Ex: ${}^nC_1 + {}^nC_3 + {}^nC_5 + \cdots = 2^{n-1}$.

Grade - Medium
Practice map, main concept: Sheet: J-M 5, 9; C-II 17 | DPP: 4(11) | Question Bank: 276, 279
Also used in: Sheet: J-I 34; C-II 13 | Question Bank: 273

Question 47
When to apply the Friedrich Karl Gauss technique and how to apply it?

Model Answer
When terms equally distant from the ends pair up (${}^nC_r = {}^nC_{n-r}$): write the sum forward and in reverse order, add, so that $2S = \sum(t_r + t_{n-r})$.
Ex: $S = \sum r\,{}^nC_r$: $2S = n\sum {}^nC_r = n2^n \Rightarrow S = n2^{n-1}$.

Grade - Low
Practice map, main concept: Sheet: J-I 30; J-II 21
Also used in: none

Question 48
When the general term contains r (type 1), how to remove r?

Model Answer

$$r\,{}^nC_r = n\,{}^{n-1}C_{r-1}, \qquad r(r-1)\,{}^nC_r = n(n-1)\,{}^{n-2}C_{r-2}$$

Ex: $\sum_{r=1}^{n} r\,{}^nC_r = n\sum {}^{n-1}C_{r-1} = n2^{n-1}$.

Grade - Medium
Practice map, main concept: Sheet: J-I 25-27, 29; J-II 24, 26; J-M 2; J-A 5; C-II 12 | DPP: 4(16) | Question Bank: 270, 274
Also used in: Sheet: J-M 72 | DPP: 3(13); 4(2)

Question 49
When the general term contains r^2 (type 1), how to remove r^2?

Model Answer
Write $r^2 = r(r-1) + r$ and use the result of item 48 on each part.

$$\sum r^2\,{}^nC_r = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

Ex: $n = 10$: $\sum r^2\,{}^{10}C_r = 10 \cdot 11 \cdot 2^8 = 28160$.

Grade - Medium
Practice map, main concept: Sheet: J-M 13, 72 | DPP: 3(13); 4(6) | Question Bank: 264
Also used in: none

Summation of Series Type 4

Question 50

How and when to use differentiation to find the sum of a series of binomial coefficients?

Model Answer

When the terms carry a factor r, $r-1, \ldots$ (no denominator): differentiate $(1+x)^n$ (or $x^k(1+x)^n$) once or twice and put $x=1$ (or -1).

Ex: $(1+x)^n$ differentiated: $n(1+x)^{n-1} = \sum r\,{}^nC_r x^{r-1}$; at $x=1$: $\sum r\,{}^nC_r = n2^{n-1}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 51

How and when to use integration to find the sum of a series of binomial coefficients?

Model Answer

When the terms carry $\frac{1}{r+1}, \frac{1}{r+2}, \ldots$: integrate $(1+x)^n$ (or $x^k(1+x)^n$) from 0 to 1 (or 0 to x).

$$\sum_{r=0}^{n} \frac{{}^nC_r}{r+1} = \frac{2^{n+1}-1}{n+1}$$

Ex: $n=3$: $1+\frac{3}{2}+1+\frac{1}{4} = \frac{15}{4} = \frac{2^4-1}{4}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 52

How to convert an integration-based question to type 1?

Model Answer

Use $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$ (and similarly for $r+2$) to absorb the denominator, then sum as in type 1.

Ex: $\sum_{r=0}^{n} \frac{{}^nC_r}{r+1} = \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1} = \frac{2^{n+1}-1}{n+1}$.

Grade - Medium

Practice map, main concept: Sheet: J-M 54, 57; C-II 5 | DPP: 3(14) | Question Bank: 288

Also used in: none

Summation of Series Type 5

Question 53

How to identify whether a question is of type 5?

Model Answer

Each term is a product ${}^nC_r \cdot {}^mC_s$ (or a sum of squares) in which $r+s$ is constant (sum pattern) or $r-s$ is constant (difference pattern).

Ex: ${}^{10}C_0\,{}^{15}C_{12} + {}^{10}C_1\,{}^{15}C_{11} + \ldots$ has lower suffixes adding to 12: sum pattern.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 54
How to use the trick in "sum = constant" pattern questions of type 5?

Model Answer

$$\sum_r {}^nC_r \, {}^mC_{k-r} = {}^{n+m}C_k$$

Ex: ${}^{10}C_0 {}^{15}C_{12} + {}^{10}C_1 {}^{15}C_{11} + \cdots + {}^{10}C_{10} {}^{15}C_2 = {}^{25}C_{12}$; $\sum({}^nC_r)^2 = {}^{2n}C_n$.

Grade - **Medium**

Practice map, main concept: Sheet: J-I 35; J-II 20, 25, 30, 40; J-M 1; J-A 6; C-II 6-8 | DPP: 4(3-4, 14-15)
Also used in: Sheet: J-II 37; J-M 2; J-A 5 | Question Bank: 274

Question 55
How to use the trick in "difference = constant" pattern questions of type 5?

Model Answer

$$\sum_r {}^nC_r \, {}^mC_{r+k} = {}^{n+m}C_{n+k}$$

Ex: $\sum {}^nC_r \, {}^nC_{r+1} = {}^{2n}C_{n+1}$.

Grade - **Low**

Practice map, main concept: Sheet: C-I 11
Also used in: none

Question 56
How to do "sum = constant" pattern questions of type 5 subjectively?

Model Answer
Multiply $(1+x)^n = \sum {}^nC_r x^r$ and $(1+x)^m = \sum {}^mC_s x^s$ and compare the coefficient of x^k with that in $(1+x)^{n+m}$.
Ex: $(1+x)^n(x+1)^n$: coefficient of x^n gives $\sum({}^nC_r)^2 = {}^{2n}C_n$.

Grade - **Low**

Practice map, main concept: none
Also used in: none

Question 57
How to do "difference = constant" pattern questions of type 5 subjectively?

Model Answer
Write one expansion in reverse order of powers, e.g. $(x+1)^m = \sum {}^mC_s x^{m-s}$, multiply with $(1+x)^n$ and compare the suitable coefficient of $(1+x)^{n+m}$.
Ex: $\sum {}^nC_r {}^nC_{r+1}$: powers $r + (n-r-1) = n-1$, so it equals the coefficient of x^{n-1} in $(1+x)^{2n}$, i.e. ${}^{2n}C_{n-1}$.

Grade - **Low**

Practice map, main concept: none
Also used in: none

Miscellaneous Questions on Summation of Series

Question 58

Observation of a GP in series questions.

Model Answer

Terms like $(1+x)^m + x(1+x)^{m-1} + \cdots + x^m$ or $(1+x)^k,\ k = r..m$ form a GP: sum with ratio $\frac{x}{1+x}$ (or $1+x$), then find the required coefficient.

Ex: $(1+x)^5 + x(1+x)^4 + \cdots + x^5 = (1+x)^6 - x^6$, so the coefficient of x^3 is ${}^6C_3 = 20$.

Grade - Medium

Practice map, main concept: Sheet: J-I 12; J-II 5, 18; J-M 7, 59, 66; C-I 12; C-II 1 | DPP: 1(2, 6) | Question Bank: 277

Also used in: Sheet: J-II 19; J-A 4 | DPP: 2(9); 4(5, 9) | Question Bank: 266, 289, 294

Question 59

Use of the properties of ${}^nC_r$ in series questions.

Model Answer

Use ${}^nC_r = {}^nC_{n-r}$ to pair terms or to write half of a symmetric sum; for odd n: ${}^nC_0 + \cdots + {}^nC_{\frac{n-1}{2}} = 2^{n-1}$.

Ex: $\sum_{r=1}^{10} {}^{21}C_r = 2^{20} - 1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 33; J-II 23; C-I 6

Also used in: Sheet: J-II 21, 25; J-M 9 | Question Bank: 270, 279, 290

Question 60

How to write an expansion with coefficients in reverse order of powers of x?

Model Answer

$(1+x)^n = C_0 + C_1x + \cdots + C_nx^n$ and $(x+1)^n = C_0x^n + C_1x^{n-1} + \cdots + C_n$ are the same polynomial with coefficients reversed; use this to multiply and compare coefficients.

Ex: $(1+x)^2 = 1 + 2x + x^2$ and $(x+1)^2 = x^2 + 2x + 1$.

Grade - Low

Practice map, main concept: none

Also used in: none

Binomial Theorem for any Index

Question 61

What are the conditions on n and x for the expansion of $(1+x)^n$ if n is not a natural number?

Model Answer

n rational (negative or fractional) and $|x| < 1$; the expansion is then an infinite series.

Ex: $(3+2x)^{-3} = 3^{-3}\left(1 + \dfrac{2x}{3}\right)^{-3}$ is valid for $|x| < \dfrac{3}{2}$.

Grade - Low

Practice map, main concept: Sheet: J-I 36

Also used in: Sheet: J-II 37

Question 62
Write all 4 expansion results.

Model Answer

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots, \qquad (1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \ldots$$

$$(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \ldots, \qquad (1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \ldots \quad (|x| < 1)$$

Ex: $(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots$; at $x = \frac{1}{2}$: $1 + \frac{2}{2} + \frac{3}{4} + \cdots = 4$.

Grade - Low
Practice map, main concept: Sheet: J-II 37
Also used in: none

Question 63
What is the coefficient of x^r in $(1-x)^{-n}$?

Model Answer

$$\text{coeff. of } x^r \text{ in } (1-x)^{-n} = {}^{n+r-1}C_r \qquad \left(\text{in } (1+x)^{-n} : \ (-1)^r\, {}^{n+r-1}C_r\right)$$

Ex: coefficient of x^5 in $(1-x)^{-3}$ is ${}^7C_5 = 21$.

Grade - Medium
Practice map, main concept: none
Also used in: DPP: 4(17) | Question Bank: 259

Question 64
How to find the coefficient of a power of x in a product of expansions when one of them is $(1-x)^{-n}$?

Model Answer
Expand the other (polynomial) factor, multiply each of its terms ax^k by the coefficient of x^{m-k} in $(1-x)^{-n}$ and add.
Ex: x^4 in $(1-x^2)(1-x)^{-3}$: ${}^6C_4 - {}^4C_2 = 15 - 6 = 9$.

Grade - Medium
Practice map, main concept: DPP: 4(9, 17) | Question Bank: 259
Also used in: none

Question 65
When can we cancel a term for simplification while finding the coefficient of a power of x in a product of two or more expansions?

Model Answer
A factor like $(1-x^k)$ can be replaced by 1 when k exceeds the required power; common factors of numerator and denominator cancel, e.g. $\dfrac{1-x^k}{1-x} = 1 + x + \cdots + x^{k-1}$.
Ex: x^4 in $\dfrac{(1-x^5)^2}{(1-x)^3}$ equals x^4 in $(1-x)^{-3}$: ${}^6C_4 = 15$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 66

Finding the coefficient of a power of x in expansions expressed in the coefficient method.

Model Answer

Write each factor as a series such as $1 + x + x^2 + \cdots = (1 - x)^{-1}$ and use ${}^{n+r-1}C_r$; this counts non-negative solutions of $a + b + \cdots = r$.

Ex: x^{10} in $(1 + x + x^2 + \dots)^3 = (1 - x)^{-3}$: ${}^{12}C_2 = 66$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 67

How to use the binomial theorem for approximations and bounds (small x, large powers)?

Model Answer

For small $|x|$: $(1 + x)^n \approx 1 + nx$ (add $\frac{n(n-1)}{2}x^2$ if needed); for bounds keep the first few positive terms, e.g. $(1 + \frac{1}{m})^m \geq 1 + 1 + \dots$.

Ex: $(1.01)^{10} \approx 1 + 10(0.01) = 1.1$; $\dfrac{1}{a + kb} \approx \dfrac{1}{a}\left(1 - \dfrac{kb}{a}\right)$ for small b.

Grade - Low

Practice map, main concept: none

Also used in: none

Miscellaneous

Question 68

Write the expansions of e^x and $\ln(1 + x)$.

Model Answer

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \qquad \ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad (-1 < x \leq 1)$$

Ex: $e = \sum \dfrac{1}{n!}$, $\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \dots$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-M 10, 19, 70 | Question Bank: 280, 299

Question 69

How to find the sum of a series based on the expansion of e^x, $\ln(1 + x)$?

Model Answer

Write the general term as (polynomial in n)$/n!$, break it into $\frac{n(n-1)}{n!} = \frac{1}{(n-2)!}$ and similar pieces, and sum each piece as e^x (for ln, match the sign and denominator with x^n/n).

Ex: $\sum_{n=1}^{\infty} \dfrac{n^2}{n!} = \sum \dfrac{n(n-1) + n}{n!} = e + e = 2e$.

Grade - Medium

Practice map, main concept: Sheet: J-M 10, 19, 68, 70 | Question Bank: 280, 299

Also used in: none